AP Statistics

Daily AI Usage and Student GPA

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Literature Review

- **Lund and Wang (2025):** This study establishes a negative correlation between high-frequency AI use and student GPAs, directly supporting our hypothesis that increased usage hours (x) result in a lower unweighted GPA (y).
- **Stanford SCALE Initiative (2025):** This report finds that heavy reliance on AI leads to "cognitive offloading," which explains why higher daily usage (x) would decrease the independent mastery needed to maintain a high GPA (y).

Why This Matters:

01 AI tools, like Chat GPT, are widely used by students

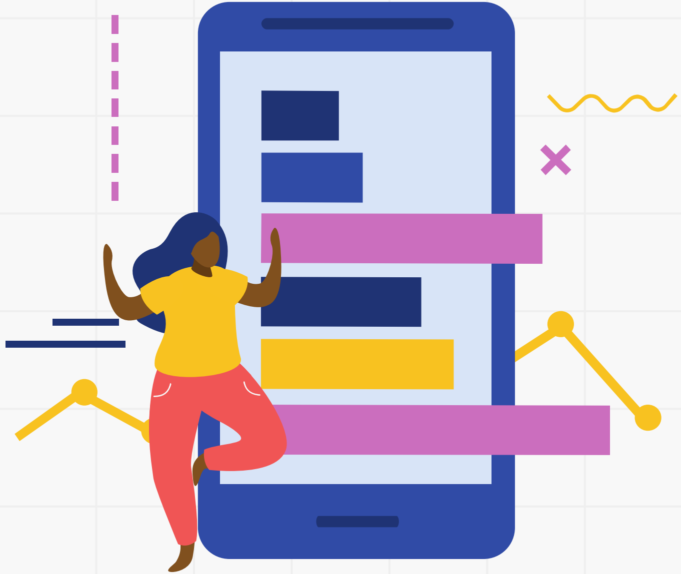
02 AI could support learning, but if used incorrectly, it could replace critical thinking

03 Most schools are concerned about academic integrity

04 Understanding this relationship helps guide responsible AI use

Research Question and Hypotheses

- **Is there a relationship between AI usage and (unweighted) GPAs ?**
  - Our explanatory variable (x) is AI usage in hours per day
  - Our response variable (y) is Doral Academy High School Students’ GPA (unweighted)
- **Hypotheses:**
  - H_o: β = 0 (*No* linear relationship)
  - H_a: β < 0 (There *is* a linear relationship)



Sampling & Data Collection

- **Population:** ~4,000 students at Doral Academy High School
- **Sample:** n= 50 students from Doral Academy High School
- **Sampling method:** We walked around during all lunches and told every 5th person we saw to fill out the survey.

Survey: AI Usage and GPA Among High School Students

This survey is anonymous and for a statistics project. Please answer honestly.

What grade are you in? (9, 10, 11, 12)

What is your current unweighted GPA? ______

On average, how many minutes per day do you use AI tools? ______

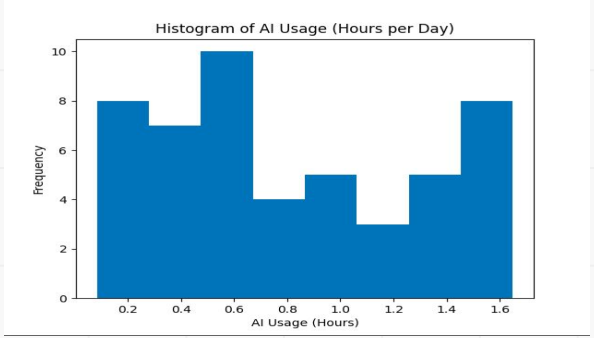
On average, how many days per week do you use AI tools? ______

Of the total AI time you use each week, about what percent is for school-related purposes? ______%

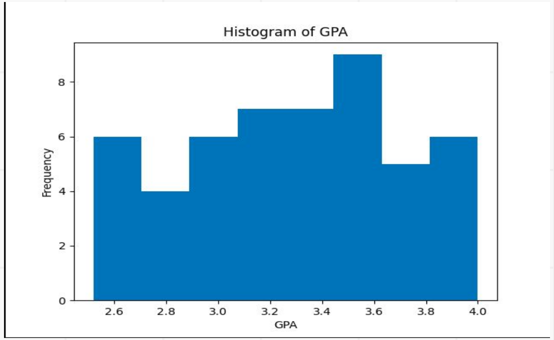
How long have you been using AI tools? (Less than 1 month, 1–3 months, 4–6 months, 7–12 months, More than 1 year)

Descriptive Statistics

- **AI Usage:**
  - **Mean**: 0.815 hours (48.9 minutes)
  - **SD**: 0.499
  - **Right-skewed**

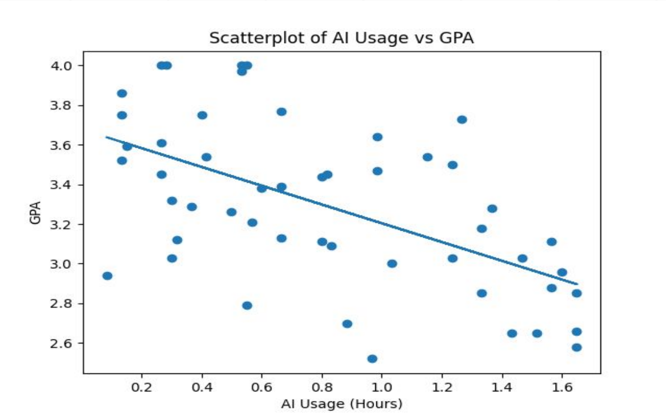


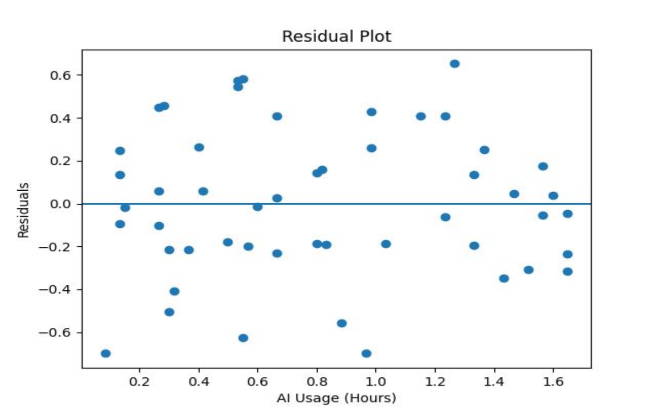
- **GPA:**
  - **Mean**: 3.291
  - **SD**: 0.416
  - **Approximately Symmetric**



Variable	Mean	SD	Min	Q1	Median	Q3	Max
AI Usage	0.82	0.50	0.08	0.33	0.73	1.26	1.65
GPA	3.29	0.42	2.52	3.01	3.29	3.58	4.00

Relationship between AI Usage and GPA





➔ Because the distribution is **Negative**, **Linear**, and **Moderately Strong**, this indicates that more AI usage actually resulted in lower GPA

- **Least-Squares Regression Line:**
  - (predicted y) = 3.677 - 0.474 (hours of AI)
  - Slope: -0.474
  - GPA decreases -.0474 per every hour of AI
  - Y-Intercept: 3.677 (Base unweighted GPA for no hours of AI)
- **Correlation:**
  - r = -0.569
  - Moderate strong, negative relationship
  - r^2 = 0.323
    - 32.2% of variation explained

Calculations

- SE = 0.099
- t = -4.79
- Df = 48
- p-value= 0.0000083

Communicate:

T-test for the slope of a least-squares regression line

β = the true slope of the regression line relating daily AI usage (in hours) to GPA for all students in Doral Academy.

$H_0: \beta = 0$ $H_a: \beta < 0$

Conditions:

- Linear: scatterplot shows a roughly linear negative pattern.
- Independent: 50 < 10% of all the high school students in Doral Academy
- Normality: residuals look approximately normal, no strong skewness nor outliers
- Equal variance: No funnel shape
- Random: we concuded a Stratifiend random sample

Since all conditions are met, we can procceed!

Calculations:

$t = \frac{b-0}{SE_b} = \frac{-0.4735}{0.0989} \approx -4.788$

$df = n-2 = 50-2 = 48$

$P(T_{48} \leq -4.79) = P(T_{48} \geq 4.79)$

tcdf(4.79, 99999, 48) ≈ 0.0000083

$p \approx 0.0000083$

Conclude:

Since our p-value is 0.0000083 is less than our significance level of 0.05, we reject the null hypothesis. Therefore, there is strong evidence statistically that there is a negative linear relationship between AI usage and GPA.

Comparison to Stanford SCALE Initiative (2025)

- **Stanford SCALE Initiative (2025):**
  - Found AI usage may reduce independent thinking
  - “Cognitive burden shift”

- **Our Study:**
  - $r = -0.569$ (moderate, negative relationship)
  - Our study adds regression + hypothesis test

- **Similarities Summarized:**
  - Similarly to our study, the Stanford SCALE Initiative done in 2025 suggested that higher AI usage is linked to lower academic performance

Limitations

- It is an Observational study, meaning we only see association, not any causation
- Responses are self-reported
- Confounding variables that can also affect gpa like:
    - Sleep
    - Study time
    - Extracurriculars
    - Course Rigor

Thank you!